Generalization in ML

Given $\quad \mathcal{D} = \{x_i, y_i\}_{i=1}^{N} \sim \text{iid } \mathbb{P}_{XY}$

$$h^*(x) = \underset{h \in \mathcal{H}}{\text{argmin}} \left[\mathbb{E}_{P_{XY}} L\big(h(x), y\big) \right]$$

$$\hat{h}^*(x) = \underset{h}{\text{argmin}} \ \frac{1}{N} \sum_{i=1}^{N} L\big(h(x), y\big)$$

$$D_{train} = \{(x_i, y_i)\}_{i=1}^{N} \sim \text{iid } \mathbb{P}_{xy}$$

$$D_{test} = \{(x_i', y_i')\}_{i=1}^{N'} \sim \text{iid } \mathbb{P}_{xy}$$

perform ERM on D_{train}, to obtain $\hat{h}^*(x)$

question: How would $\hat{h}^*(x)$ perform on D_{test}?

Consider $L : h(x) \times Y \longrightarrow \mathbb{R}^+$

$$L(\quad) = \|h(x) - y\|_2^2$$

$$x \in \mathbb{R}^d \qquad y \in \mathbb{R}$$

$$R(h) = \mathbb{E}_{\mathbb{P}_{xy}}\left[\|h(x) - y\|_2^2\right]$$

$$h^*(x) = \underset{h}{\operatorname{argmin}}\left[R(h)\right]$$

$$= \mathbb{E}(y|x).$$

$$= \int_Y y \, p_{y|x}(y|x) \, dy.$$

Define $h_D(x)$ to be a hypothesis "learned" using a dataset D.

↳ solve ERM

Let P_D be the distribution according to which we sample a data set from $\mathbb{P}_{xy}$.

Define an "average" hypothesis as

$$\bar{h}(x) = \mathbb{E}_{P_D}\left[h_D(x)\right]$$

Consider average Risk on all dataset:

$$R(h) = \mathbb{E}_{P_D}\left[R(h_D)\right]$$

$$= \mathbb{E}_{P_D}\mathbb{E}_{P_{xy}}\left[\left\| h_D(x) - y \right\|_2^2\right]$$

$$= \mathbb{E}_{P_D}\mathbb{E}_{P_{xy}}\left[\left\| h_D(x) - \bar{h}(x) + \bar{h}(x) - y \right\|_2^2\right]$$

$$= \mathbb{E}_{P_D}\mathbb{E}_{P_{xy}}\left[\left(h_D(x) - \bar{h}(x)\right)^2\right] + \mathbb{E}_{P_D}\mathbb{E}_{P_{xy}}\left(\bar{h}(x) - y\right)^2$$

$$+ \underbrace{2 \cdot \mathbb{E}_{P_D}\mathbb{E}_{P_{xy}}\left(h_D(x) - \bar{h}(x)\right)\left(\bar{h}(x) - y\right)}_{=0}$$

$$R(h) = \mathbb{E}_{P_D} \mathbb{E}_{P_{xy}} \left[\left(h_D(x) - \bar{h}(x) \right)^2 \right] + \mathbb{E}_{P_{xy}} \left(\bar{h}(x) - y \right)^2$$

Consider

$$\mathbb{E}_{P_{xy}} \left[\left(\bar{h}(x) - y \right)^2 \right] =$$

$$\mathbb{E}_{P_{xy}} \left[\left(\bar{h}(x) - h^*(x) + h^*(x) - y \right)^2 \right]$$

$$= \mathbb{E}_{P_{xy}} \left[\left(h^*(x) - y \right)^2 \right] + \mathbb{E}_{P_{xy}} \left(\bar{h}(x) - h^*(x) \right)^2$$

$$+ \underbrace{2 \, \mathbb{E}_{P_{xy}} \left(\bar{h}(x) - h^*(x) \right) \left(h^*(x) - y \right)}_{= 0}$$

Finally,

$$R(h) = \overbrace{\mathbb{E}_{P_D} \mathbb{E}_{P_{xy}} \left[h_D(x) - \bar{h}(x) \right]^2}^{\text{variance of } h} + \overbrace{\mathbb{E}_{P_{xy}} \left[\bar{h}(x) - h^*(x) \right]^2}^{\text{Bias}}$$

$$+ \underbrace{\mathbb{E}_{P_{xy}} \left[h^*(x) - y \right]^2}_{\text{Irreducible "noise"}}$$

Analysis of Bias & variance.

Consider the Bias term:

$\mathbb{E}\left[\bar{h}(x) - h^*(x)\right]$: quantifies the effect of a particular choice of hypothesis class.

Variance:

$\mathbb{E}_{P_D}\mathbb{E}_{P_{xy}}\left[h_D(x) - \bar{h}(x)\right]^2$: measures the sensitivity of the hypothesis to the choice of dataset.

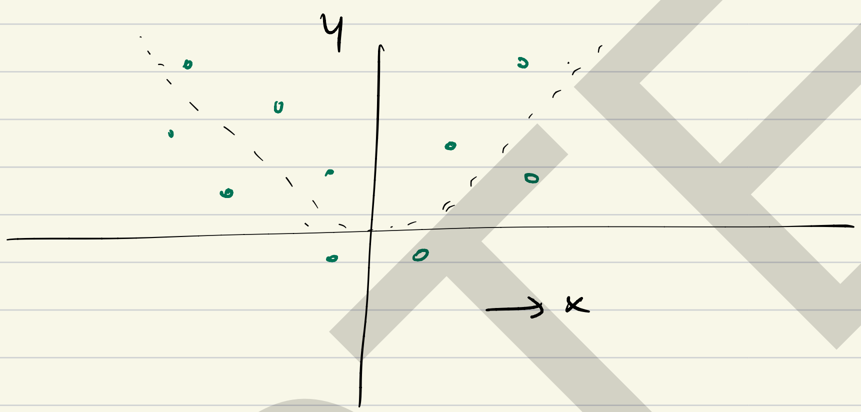
Bias & variance in practice.

Example:

$x \in \mathbb{R}, \quad y \in \mathbb{R},$

$y = ax^2 + b + \epsilon \qquad \epsilon \sim N(0, I)$

$a, b \in \mathbb{R}^+$



1) $h_1(x) = 2 \quad \forall \quad x$

Bias : High

var : 0

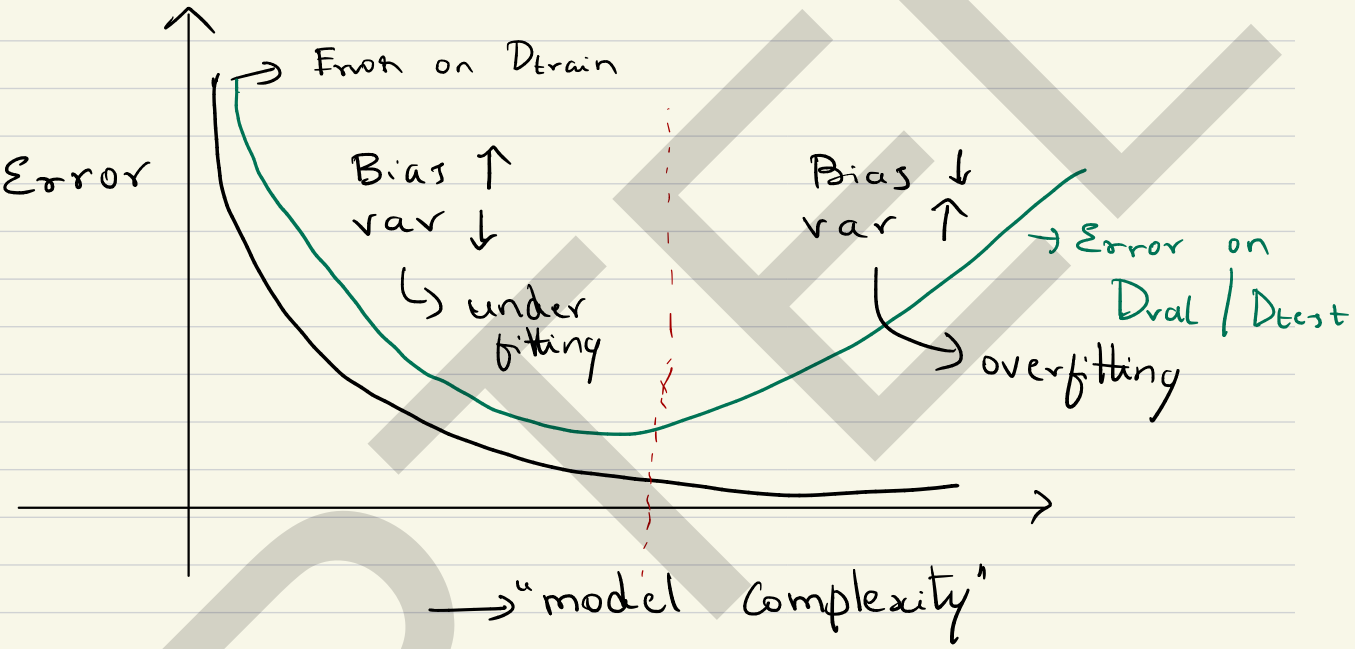
2) $h_2(x_i) = y_i \quad \forall \quad x_i$ in D : lookup table.

Bias : 0 , $h_2(x_i) = 0$ otherwise

var : high.

Pragmatic way of trading B & V

Typically, in a small-sample setting the following behavior is observed.



Tip: Always start with a hypothesis that operates in the low bias & high var (overfit) region via overparameterization.